		TL;DR		UF-P-2		UF-P-4		PersonalLLM	
		ALL	AVG	ALL	AVG	ALL	AVG	ALL	AVG
Seen	G-Oracle	$73.06_{\pm 0.23}$	$73.06_{\pm 0.23}$	$64.53_{\pm 0.14}$	$64.53_{\pm 0.14}$	$61.52_{\pm 0.13}$	$61.52_{\pm 0.13}$	N/A	N/A
	Uniform	$49.62_{\pm 0.09}$	$49.62_{\pm 0.09}$	$61.82_{\pm 0.16}$	$61.82_{\pm 0.16}$	$56.15_{\pm 0.22}$	$56.15_{\pm 0.22}$	$62.91_{\pm 0.07}$	$62.91_{\pm 0.07}$
	I2E	$49.93_{\pm 0.23}$	$49.74_{\pm 0.06}$	$61.48_{\pm 0.18}$	$61.49_{\pm 0.70}$	$57.21_{\pm 0.37}$	$57.44_{\pm 0.37}$	$65.74_{\pm 0.04}$	$65.77_{\pm 0.05}$
	$\text{I2E}_{\text{proxy}}$	$49.80_{\pm 0.16}$	$49.54_{\pm 0.13}$	$61.43_{\pm 0.56}$	$61.33_{\pm 0.61}$	$56.78_{\pm 0.14}$	$57.14_{\pm 0.31}$	$65.66_{\pm 0.11}$	$65.77_{\pm 0.05}$
	VPL	$49.52_{\pm 0.14}$	$49.44_{\pm 0.21}$	$61.11_{\pm 0.16}$	$61.86_{\pm 0.84}$	$56.04_{\pm 1.71}$	$56.77_{\pm 0.38}$	$70.84_{\pm 0.18}$	$67.95_{\pm 0.21}$
	PAL	$50.12_{\pm 0.13}$	$50.15_{\pm 0.15}$	$59.95_{\pm 0.04}$	$61.53_{\pm 0.22}$	$56.95_{\pm 0.13}$	$57.37_{\pm 0.14}$	$66.25_{\pm 0.35}$	$66.29_{\pm 0.06}$
	CoPL	$\mathbf{96.58}_{\pm 0.09}$	$\mathbf{96.19}_{\pm 0.02}$	$\mathbf{63.81}_{\pm 0.16}$	$\mathbf{63.45}_{\pm 0.38}$	$\mathbf{62.57}_{\pm 0.38}$	$\mathbf{62.08}_{\pm 0.27}$	$\mathbf{74.85}_{\pm 0.17}$	$\mathbf{74.37}_{\pm 0.03}$
Unseen	G-Oracle	$72.55_{\pm 1.79}$	$72.55_{\pm 1.79}$	$64.66_{\pm 1.10}$	$64.66_{\pm 1.10}$	$61.33_{\pm 0.35}$	$61.33_{\pm 0.35}$	N/A	N/A
	Uniform	$50.11_{\pm 0.36}$	$50.11_{\pm 0.36}$	$62.82_{\pm 0.59}$	$62.82_{\pm 0.59}$	$55.65_{\pm 0.61}$	$55.65_{\pm 0.61}$	$62.97_{\pm 0.07}$	$62.97_{\pm 0.07}$
	I2E	$49.85_{\pm 0.38}$	$49.16_{\pm 0.82}$	$61.67_{\pm 0.82}$	$59.52_{\pm 0.51}$	$56.42_{\pm 0.41}$	$56.75_{\pm 0.68}$	$65.79_{\pm 0.18}$	$66.11_{\pm 0.24}$
	$\text{I2E}_{\text{proxy}}$	$49.75_{\pm 0.94}$	$49.12_{\pm 0.57}$	$62.30_{\pm 0.54}$	$61.70_{\pm 0.63}$	$56.00_{\pm 1.15}$	$56.50_{\pm 0.34}$	$65.49_{\pm 0.10}$	$65.79_{\pm 0.04}$
	VPL	$49.40_{\pm 0.88}$	$49.31_{\pm 0.57}$	$60.83_{\pm 0.40}$	$62.62_{\pm 0.49}$	$54.03_{\pm 1.54}$	$56.13_{\pm 0.57}$	$71.31_{\pm 0.58}$	$68.55_{\pm 0.47}$
	PAL	$49.48_{\pm 0.86}$	$49.64_{\pm 0.55}$	$59.83_{\pm 0.69}$	$61.71_{\pm 0.31}$	$57.07_{\pm 0.22}$	$57.13_{\pm 0.33}$	$65.94_{\pm 0.11}$	$66.40_{\pm 0.03}$
	CoPL	$\mathbf{96.71}_{\pm 0.25}$	$\mathbf{96.21}_{\pm 0.14}$	$\mathbf{63.92}_{\pm 0.54}$	$\mathbf{63.26}_{\pm 0.51}$	$\mathbf{61.62}_{\pm 0.10}$	$\mathbf{61.97}_{\pm 0.35}$	$\mathbf{75.69}_{\pm 0.22}$	$\mathbf{75.49}_{\pm 0.03}$

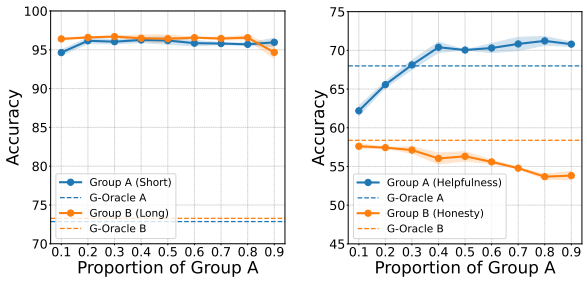
Table 2: Accuracy of reward models on unseen annotated pairs. The results report performance on *Seen users* encountered during training and on *Unseen users*. **Bold** represents the best result, except for G-Oracle. These results are based on `gemma-2b-it`. Additional results using `gemma-7b-it` and `gemma2-27b-it` are represented in Table A1 and Table A3, respectively.

	G-Oracle	Uniform	I2E	$\text{I2E}_{\text{proxy}}$	VPL	PAL	CoPL
Common	$71.86_{\pm 0.14}$	$\mathbf{74.52}_{\pm 0.45}$	$73.94_{\pm 0.21}$	$74.15_{\pm 1.53}$	$72.73_{\pm 1.00}$	$70.82_{\pm 0.17}$	$71.23_{\pm 1.63}$
Controversial	$57.68_{\pm 0.27}$	$49.86_{\pm 0.30}$	$49.61_{\pm 0.05}$	$49.86_{\pm 0.06}$	$50.26_{\pm 0.44}$	$49.79_{\pm 0.12}$	$\mathbf{56.89}_{\pm 1.56}$
Total	$64.53_{\pm 0.14}$	$61.82_{\pm 0.16}$	$61.48._{\pm 0.18}$	$61.59_{\pm 0.79}$	$61.11_{\pm 0.32}$	$59.95_{\pm 0.04}$	$\mathbf{63.81}_{\pm 0.15}$

Table 3: Accuracy of reward models on UF-P-2 (ALL) with `gemma-2b-it`, broken down by pair type. *Common* refers to pairs for which the two preference groups provide the same preference label, *Controversial* refers to pairs labeled differently by the two groups, and *Total* encompasses all pairs. These categories reflect how diverse user preferences affect the performance of reward models. **Bold** represents the best result, except with G-Oracle.

G-Oracle on controversial pairs while preserving high accuracy on common pairs.

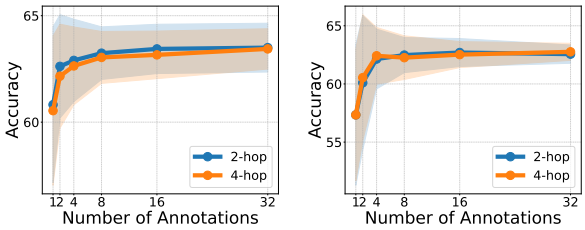
Performance under imbalanced group distributions. We vary the group proportion from 1:9 to 9:1 on the TL;DR (AVG) and UF-P-2 (AVG) datasets. As shown in Fig. 4, CoPL consistently captures both majority and minority preferences, maintaining stable accuracy for the short- and long-summary groups on TL;DR. On UF-P-2, CoPL still reflects diverse preferences, but the gap relative to the balanced 5:5 setting widens as the ratio becomes more skewed. Majority accuracy rises while minority accuracy falls, showing majority bias under imbalance. The lower absolute accuracy for the honesty group reflects the inherent difficulty of that preference, which remains consistent with the G-oracle results. The difference between TL;DR and UF-P-2 is also explained by UF-P-2 containing common pairs on which both groups agree, which reduces the distinct signal from the minority.



(a) TL;DR (AVG) (b) UF-P-2 (AVG)

Figure 4: Group-wise accuracy of reward models with Gemma-2b-it in TL;DR (AVG) and UF-P-2 (AVG), varying the ratio of group size (A:B) from 1:9 to 9:1 with the total number of users fixed at 10,000.

Figs. A4 and A5 show the learned user embeddings for TL;DR and UF-P-2. CoPL preserves well-separated clusters aligned with group identities even under extreme imbalance. Thus, while minority group accuracy may drop, the representation space remains robust to group structure. To miti-



(a) UF-P-2 (AVG) (b) UF-P-4 (AVG)

Figure 5: Accuracy of unseen user adaptation as the number of provided annotation sets increases, evaluated on UF-P-2/4 (AVG) with `gemma-2b-it`. *2-hop* and *4-hop* indicates 2-hop and 4-hop adaptation, respectively.

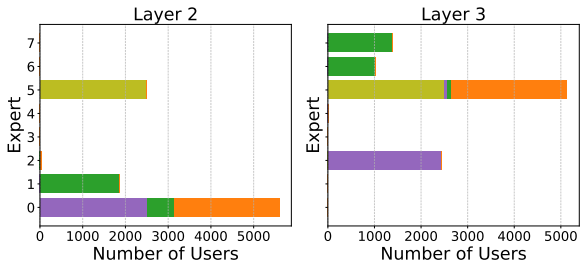


Figure 6: Expert allocation at layers 2 and 3 in UF-P-4 (ALL) with `gemma-2b-it`. Colors indicate preference groups. Users with similar preference groups are mapped to the same expert.

gate majority bias, we can apply loss reweighting, such as focal loss (Lin et al., 2017; Subramanian et al., 2021), to emphasize underrepresented groups in the reward model training.

In the four-group setting, additional UF-P-4 (AVG) results exhibit the same trend, as reported in Section E.

Effect of the number of annotations in unseen user adaptation. Fig. 5 shows accuracy as the number of provided annotations increases in UF-P-2 (AVG) and UF-P-4 (AVG). We observe that additional annotations lead to more accurate preference predictions for unseen users in general. However, in practice, even eight annotations are sufficient, enabling accurate inference of each user's preference. We also compare two-hop and four-hop adaptations, but there is no significant difference.

Ablation study of CoPL. Table 4 presents an ablation study of CoPL, focusing on GNN-derived user embeddings and the MoLE architecture. When GNN embeddings are removed, user representations become learnable parameters. Without MoLE, user embeddings are projected into the token space and passed as an additional token to the reward model. The results indicate that components of

	UF-P-2 (ALL)	UF-P-4 (ALL)
CoPL	**63.81**$_{\pm 0.16}$	**62.57**$_{\pm 0.38}$
w/o GNN embedding	62.09$_{\pm 0.38}$	56.75$_{\pm 0.30}$
w/o MoLE ($n = 64$)	62.69$_{\pm 0.86}$	62.28$_{\pm 0.33}$
w/o MoLE ($n = 16$)	62.43$_{\pm 0.69}$	62.13$_{\pm 0.12}$

Table 4: Ablation study of CoPL in UF-P-2/4 (ALL) with `gemma-2b-it`. *w/o GNN embedding* replaces user embeddings from GNN with learnable user embeddings. *w/o MoLE* removes the MoLE and projects user embeddings into the token space. The symbol n denotes the LoRA rank.

	UF-P-4 (ALL)	UF-P-4 (AVG)
CoPL	**61.62**$_{\pm 0.10}$	**61.97**$_{\pm 0.35}$
Naive Avg.	59.91$_{\pm 0.59}$	59.39$_{\pm 0.50}$
User Opt.	59.24$_{\pm 0.71}$	59.45$_{\pm 0.72}$

Table 5: Accuracy of unseen-user adaptation in UF-P-4 (ALL/AVG) with `gemma-2b-it`. *Naive Avg.* computes the unseen user's embedding as the unweighted average of 2-hop neighbors, while CoPL applies a weighted average. *User Opt.* represents an optimization-based approach that learns a parameterized user embedding by maximizing the likelihood of the given annotations.

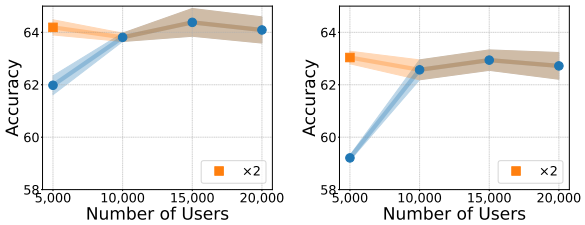
CoPL are effective. Specifically, GNN-based embeddings are a crucial component of CoPL, and the MoLE architecture further enhances accuracy. Notably, CoPL uses fewer activated parameters than w/o MoLE ($n = 64$).

Fig. 6 depicts expert allocation across layers two and three, where the user-conditioned gating mechanism partitions users differently at each layer. We can observe that users with the same preferences tend to be routed to the same expert.

We provide the ablation study of the number of experts in Section E.

Ablation study of unseen user adaptation. We conduct an ablation study to evaluate the effectiveness of the unseen user adaptation strategy, comparing it to two baselines, Naive Avg and User Opt. Naive Avg assigns each unseen user embedding as the unweighted average of 2-hop seen user embeddings. User Opt replaces $\boldsymbol{e}_u^{(L)}$ with a parameterized embedding learned by minimizing Equation (5) on the provided annotations. Table 5 reports results in UF-P-4-ALL/AVG with `gemma-2b-it`, showing that CoPL outperforms both alternatives while achieving better computational efficiency than the optimization-based User Opt.

Fig. A2 illustrates that naive averaging places



(a) UF-P-2 (ALL) (b) UF-P-4 (ALL)

Figure 7: Accuracy of reward models on UF-P-2 and UF-P-4 (ALL) with `gemma-2b-it` with varying number of seen users. The number of annotations per user remains constant except in the case with "$\times 2$," where we double the per-user annotations only for $5,000$ users, making the total number of annotations $10,000$.

unseen users away from identical preference group users, whereas our method clusters them more closely with users who share the same preferences.

Ablation study of the number of users. We conduct an ablation study of CoPL by varying the number of users and report the performance in Fig. 7. The performance of the model is consistent except for the case where there are only $5,000$ users in the training set. The performance with 5,000 users becomes comparable when we double the number of annotations $(2\times)$, indicating the need for a sufficient amount of annotations to capture diverse preferences.

Training reward models with GNN. Table 6 reports GNN accuracy on seen users and responses for test pairs excluded from the training dataset. The results demonstrate that GNN can accurately predict labels for unannotated pairs with sparse annotations. We provide the additional ablation study of message-passing in Section E.

Table 7 examines the impact of training with GNN-based pseudo labels, allowing the model to leverage additional preference data. Although the pseudo-labeled pairs increase the dataset size, performance is slightly worse than using only user-provided annotations, suggesting that noise degrades model accuracy.

To investigate the effect of noise further, a user-specific reward model is trained on pseudo labels for a random sample of 10 users per group. The results are considerably worse than the G-Oracle, indicating that noisy labels introduce training instability. This observation aligns with Wang et al. (2024a), which notes that noisy preference labels can lead to training instability and performance degradation.

UF-P-2		UF-P-4	
ALL	AVG	ALL	AVG
$84.84_{\pm 0.83}$	$84.32_{\pm 0.09}$	$90.01_{\pm 0.35}$	$87.74_{\pm 0.19}$

Table 6: Test accuracy of the GNN. We evaluate the model using the same users from training but with annotation pairs that are not reflected in the graph.

	UF-P-2 (ALL)	UF-P-4 (ALL)
CoPL	$63.81_{\pm 0.16}$	$62.57_{\pm 0.38}$
Pseudo label	$62.77_{\pm 0.70}$	$62.26_{\pm 0.27}$
G-Oracle	$64.53_{\pm 0.14}$	$61.52_{\pm 0.13}$
User-specific	$58.09_{\pm 1.73}$	$55.30_{\pm 3.30}$

Table 7: Accuracy of reward model trained by using a pre-trained GNN in UF-P-2/4 (ALL) with `gemma-2b-it`. The *"pseudo-label"* trains a reward model on all seen user–response pairs, with annotations provided by GNN-predicted labels. The *"user-specific"* refers to a BTL model trained with pseudo-labels for each user. Only 10 users per group are sampled due to computational cost.

6 Conclusion

In this work, we introduced CoPL, a novel approach for personalizing LLMs through graph-based collaborative filtering and MoLE. Unlike existing methods that treat user preferences independently or require predefined clusters, our approach leverages multi-hop user-response relationships to improve preference estimation, even in sparse annotation settings. By integrating user-specific embeddings into the reward modeling process with MoLE, CoPL effectively predicts an individual preference.

Limitations

This work demonstrates how GCF-based user embeddings enable personalization in sparse settings, but we do not extensively explore other GNN architectures that could further reduce sample complexity. Additionally, although CoPL employs a gating mechanism for user-specific expert allocation, we did not apply load-balancing loss, which induces more even activation among experts. As a result, some experts remain inactive in Fig. 6. Future work may investigate different GNN designs and incorporate load-balancing techniques to fully leverage the potential of GNN and MoLE, respectively.

The group-wise oracle model may appear underwhelming, likely because our smaller backbone LLM struggles to capture subtle stylistic differences between responses. Larger-scale models